



8438 - The first spectrum of a temperate super-Jupiter planet

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Eps Ind Ab				
	4	NIRSpec	NIRSpec IFU Spectroscopy	(4) eps-Ind-Ab-2026-may
	9	NIRSpec	NIRSpec IFU Spectroscopy	(4) eps-Ind-Ab-2026-may
	3	MIRI MRS ON	MIRI Medium Resolution Spectroscopy	(1) eps-Ind-Ab
	8	MIRI MRS OFF	MIRI Medium Resolution Spectroscopy	(2) eps-Ind-Ab-BACKGROUND

ABSTRACT

Giant exoplanets, characterized by their substantial size and gaseous composition, play a crucial role in enhancing our understanding of planetary formation, evolution, and atmospheric chemistry beyond our Solar System. The aim of this proposal is to achieve the first medium-resolution spectroscopy of a temperate super-Jupiter planet, Epsilon Indi Ab. With a cool temperature of approximately 275 K and its proximity of 3.6 pc, Epsilon Indi Ab is particularly bright in the mid-infrared. Its wide separation of 4.1" from its host star presents a unique opportunity for direct spectroscopy. These distinctive features make it an ideal candidate for obtaining a medium-resolution spectrum using JWST. Results of this project will deepen our understanding of chemical enrichment processes in low-temperature, older gas giants, while also providing a valuable comparison with the gas giants found in our Solar System. Additionally, preliminary indications suggest that the atmosphere may be carbon-rich, and our spectral analysis will help to test this hypothesis.

OBSERVING DESCRIPTION

We plan to utilize the NIRSpec and MIRI-MRS instruments in IFU mode to obtain a medium-resolution spectrum of the exoplanet Epsilon Indi Ab from 2.9 to 5.3 μm in the near-infrared and 4.9 to 27.9 μm in the mid-infrared. The strategy for MIRI-MRS and NIRSpec observations uses 4-point dithers. There are possible conflicts between favoured position angles to limit the contamination by the host star in long channels of MIRI and MAZ. The observing strategy may be optimized upon request from the STScI in phase 2.

We set our goal in terms of signal-to-noise ratio (SNR) to a value of 100 at the peak for NIRSpec and 50 for MIRI, to clearly detect key molecular features within a reasonable time. To estimate this SNR, we rely on the Exposure Time Calculator (ETC) and models, which reproduce the MIRI coronagraphy fluxes.

With 2 integrations and 58 groups with SLOWR1 readout for MIRI and 1 integration and 50 groups with the NRSIRS2RAPID readout pattern for NIRSpec, our goal in terms of peak SNR is reached. The use of SLOWR1 is required to limit data rate. To ensure the maximum quality of the data and given the low flux of the source, we also include a dedicated background observation for MIRI, following the successful strategy adopted by the ERS program on direct spectroscopy of exoplanets (ERS 1386).

Proposal 8438 - Targets - The first spectrum of a temperate super-Jupiter planet

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	eps-Ind-Ab	RA: 22 03 21.7878 (330.8407825d) Dec: -56 47 6.56 (-56.78516d) Equinox: J2000	Proper Motion RA: 3966.6610000000005 mas/yr Proper Motion Dec: -2536.1920000932514 mas/yr Parallax: 0.2748431" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanets] Extended=NO				
	(2)	eps-Ind-Ab-BACKGROUND	RA: 22 03 26.5000 (330.8604167d) Dec: -56 46 0.00 (-56.76667d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				
	(3)	-eps-Ind	RA: 22 03 21.6536 (330.8402233d) Dec: -56 47 9.52 (-56.78598d) Equinox: J2000	Proper Motion RA: 3966.6610000000005 mas/yr Proper Motion Dec: -2536.1920000932514 mas/yr Parallax: 0.2748431" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars]				
	(4)	eps-Ind-Ab-2026-may	RA: 22 03 21.8818 (330.8411742d) Dec: -56 47 6.48 (-56.78513d) Equinox: J2000	Proper Motion RA: 3966.6610000000005 mas/yr Proper Motion Dec: -2536.1920000932514 mas/yr Parallax: 0.2748431" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanets] Extended=NO				

Proposal 8438 - Observation 4 - The first spectrum of a temperate super-Jupiter planet

Observation	Proposal 8438, Observation 4: NIRSpec											Mon May 04 16:00:09 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 4:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	eps-Ind-Ab-2026-may	RA: 22 03 21.8818 (330.8411742d) Dec: -56 47 6.48 (-56.78513d) Equinox: J2000				Proper Motion RA: 3966.6610000000005 mas/yr Proper Motion Dec: -2536.1920000932514 mas/yr Parallax: 0.2748431" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Exoplanets] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	25	1	false	true	NONE	4	4	1517.245	222762

Proposal 8438 - Observation 4 - The first spectrum of a temperate super-Jupiter planet

Special Requirements	Aperture PA Range 21.97164917 to 33.97164917 Degrees (V3 243.0 to 255.0)
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Proposal 8438 - Observation 9 - The first spectrum of a temperate super-Jupiter planet

Observation	Proposal 8438, Observation 9: NIRSpec											Mon May 04 16:00:09 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 9:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	eps-Ind-Ab-2026-may	RA: 22 03 21.8818 (330.8411742d) Dec: -56 47 6.48 (-56.78513d) Equinox: J2000				Proper Motion RA: 3966.6610000000005 mas/yr Proper Motion Dec: -2536.1920000932514 mas/yr Parallax: 0.2748431" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Exoplanets] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	25	1	false	true	NONE	4	4	1517.245	222762

Proposal 8438 - Observation 9 - The first spectrum of a temperate super-Jupiter planet

Special Requirements	Aperture PA Range 63.97164917 to 73.97164917 Degrees (V3 285.0 to 295.0)
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Proposal 8438 - Observation 3 - The first spectrum of a temperate super-Jupiter planet

Observation	Proposal 8438, Observation 3: MIRI MRS ON												Mon May 04 16:00:09 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Comments: This is a background observation to complement Observation 20.												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	eps-Ind-Ab	RA: 22 03 21.7878 (330.8407825d) Dec: -56 47 6.56 (-56.78516d) Equinox: J2000				Proper Motion RA: 3966.6610000000005 mas/yr Proper Motion Dec: -2536.1920000932514 mas/yr Parallax: 0.2748431" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[Exoplanets] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	3 -eps-Ind	FND	FAST	4	1		1	11.1	12227.15			
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	LONG(C)	MRSLONG		SLOWR1	58	2	1	Dither 1	4	8	11180.483	222762
	1	LONG(C)	MRSSHORT		SLOWR1	58	2	1	Dither 1	4	8	11180.483	222762
	2	MEDIUM(B)	MRSLONG		SLOWR1	58	2	1	Dither 1	4	8	11180.483	222762
	2	MEDIUM(B)	MRSSHORT		SLOWR1	58	2	1	Dither 1	4	8	11180.483	222762
	3	SHORT(A)	MRSLONG		SLOWR1	58	2	1	Dither 1	4	8	11180.483	222762
	3	SHORT(A)	MRSSHORT		SLOWR1	58	2	1	Dither 1	4	8	11180.483	222762
	3	SHORT(A)	MRSSHORT		SLOWR1	58	2	1	Dither 1	4	8	11180.483	222762

Proposal 8438 - Observation 3 - The first spectrum of a temperate super-Jupiter planet

Special Requirements	Sequence Observations 3, 8, Non-interruptible
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Proposal 8438 - Observation 8 - The first spectrum of a temperate super-Jupiter planet

Observation	Proposal 8438, Observation 8: MIRI MRS OFF												Mon May 04 16:00:09 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Comments: This is a background observation to complement Observation 20.												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	eps-Ind-Ab-BACKGROUND	RA: 22 03 26.5000 (330.8604167d) Dec: -56 46 0.00 (-56.76667d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	2-Point					BACKGROUND			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	LONG(C)	MRSLONG		SLOWR1	60	2	1	Dither 1	2	4	5781.361	222762
	1	LONG(C)	MRSSHORT		SLOWR1	60	2	1	Dither 1	2	4	5781.361	222762
	2	MEDIUM(B)	MRSLONG		SLOWR1	60	2	1	Dither 1	2	4	5781.361	222762
	2	MEDIUM(B)	MRSSHORT		SLOWR1	60	2	1	Dither 1	2	4	5781.361	222762
	3	SHORT(A)	MRSLONG		SLOWR1	60	2	1	Dither 1	2	4	5781.361	222762
	3	SHORT(A)	MRSSHORT		SLOWR1	60	2	1	Dither 1	2	4	5781.361	222762

Proposal 8438 - Observation 8 - The first spectrum of a temperate super-Jupiter planet

Special Requirements	Sequence Observations 3, 8, Non-interruptible
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